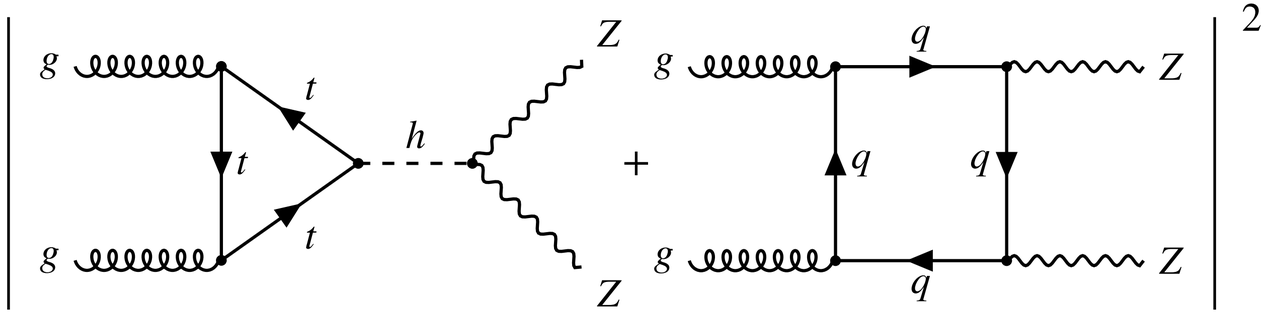
# Asimov == median of toy pseudo-experiments check for histogram & NSBI-based test statistics

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Tae Hyoun Park

# $gg \to H^* \to 4\ell$ signal probability model




- The probability density of an event according to the signal-background-interference (SBI) hypothesis from individual contributions:

$$p_{\mathrm{SBI}}(x|\mu) = \frac{\mu\nu_{\mathrm{S}}p_{\mathrm{S}}(x) + \sqrt{\mu}\nu_{\mathrm{I}}p_{\mathrm{I}}(x) + \nu_{\mathrm{B}}p_{\mathrm{B}}(x)}{\mu\nu_{\mathrm{S}} + \sqrt{\mu}\nu_{\mathrm{I}} + \nu_{\mathrm{B}}}$$

- Re-arranging because we don't want to deal with the negative interference:

$$p_{\mathrm{SBI}}(x|\mu) = \frac{(\mu - \sqrt{\mu})\nu_{\mathrm{S}}p_{\mathrm{S}}(x) + \sqrt{\mu}\nu_{\mathrm{SBI}}p_{\mathrm{SBI}}(x) + (1 - \sqrt{\mu})\nu_{\mathrm{B}}p_{\mathrm{B}}(x)}{(\mu - \sqrt{\mu})\nu_{\mathrm{S}} + \sqrt{\mu}\nu_{\mathrm{SBI}} + (1 - \sqrt{\mu})\nu_{\mathrm{B}}}$$

# Statistical analysis setup

**Histogram**

- Histogram signal, background, interference events.
- Histogram ≈ probability density.
- For $\mu \neq 1$ scenario, each histogram gets scaled by corresponding factors.
- Assume $\mu = 1$ as Asimov dataset.
- Compute likelihood = (product of Poisson probabilities per-bin)

$$\mathcal{L} = \prod_{\text{bins}} \mathcal{P}(n|\nu)$$

**NSBI**

- Choosing $p_{\text{ref}} = p_{\text{bkg}}$.
- 2 CARL NNs to estimate density ratios: $p_{\text{sig}}/p_{\text{ref}}$, $p_{\text{sbi}}/p_{\text{ref}}$
- For $\mu \neq 1$ scenario, each DRE gets scaled by corresponding factors. Estimate mixture model *ratio*, $p_{\text{SBI}}/p_{\text{bkg}}$.
- Assume $\mu = 1$ as Asimov dataset.
- Compute likelihood ratio = (product of density ratios per-event).
- The denominator cancels out in ratio of ratios.

$$\mathcal{L} = \mathcal{P}(n|\nu) \prod_{\text{events}} \frac{p_{\text{SBI}}(x|\mu)}{p_{\text{ref}}(x)}$$

- Important note: the “total rate” Poisson term, is not estimated using NNs.

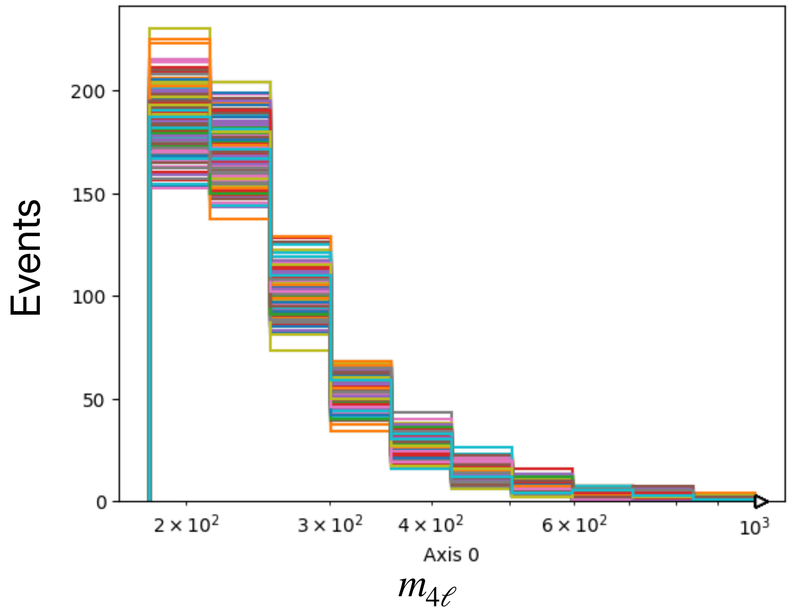
# Expected signal-background interference yields

| | 1 fib | 10 ifb | 100 fib | 1000 ifb |
|---|---|---|---|---|
| SIG | 0.310 | 3.099 | 30.990 | 309.897 |
| BKG | 5.953 | 59.530 | 595.303 | 5953.034 |
| INT | -0.606 | -6.064 | -60.643 | -606.433 |
| SBI | 5.656 | 56.565 | 565.650 | 5656.498 |

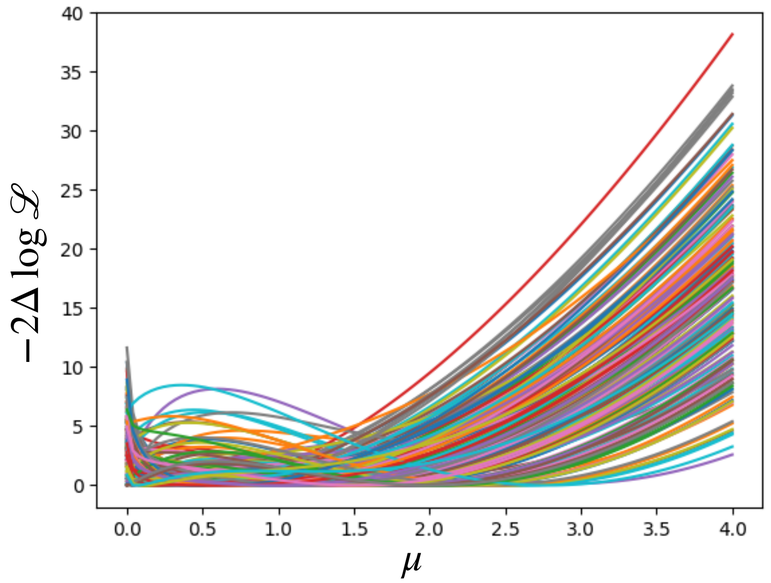
# Throwing toys

- Pseudo-experiments are thrown by generating a toy weight for each event, sampled from Poisson with mean of nominal weight.

$$w_{\text{toy}} \sim \mathscr{P}(\lambda = w)$$



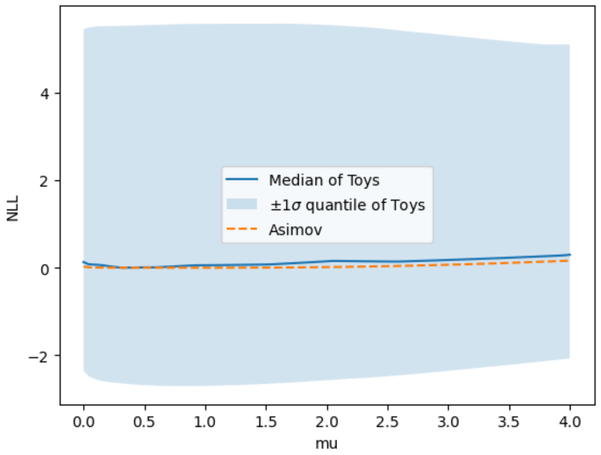

(Here, the y-axis is not consistent between toys — some toy curves should go negative w.r.t. nominal Asimov curve)



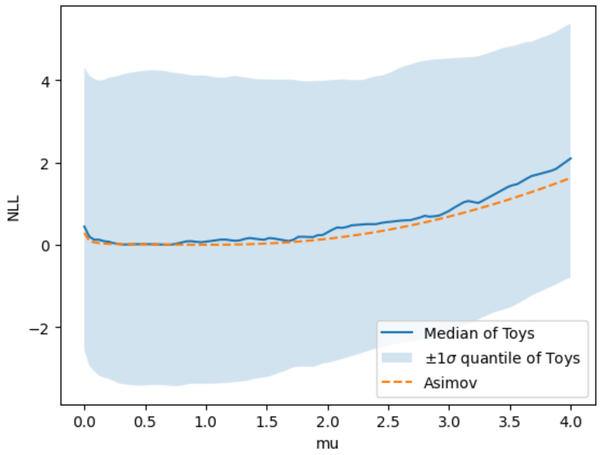

- Expectation: the NLL of nominal Asimov dataset corresponds to the median of throwing these pseudo-experiments.

# Asimov vs. Median(Toys): NSBI

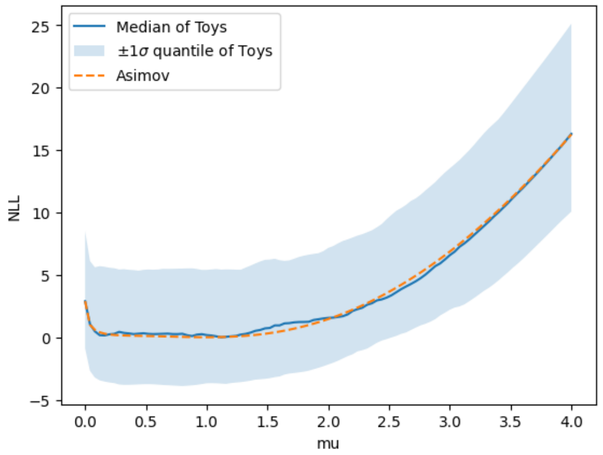
1 ifb



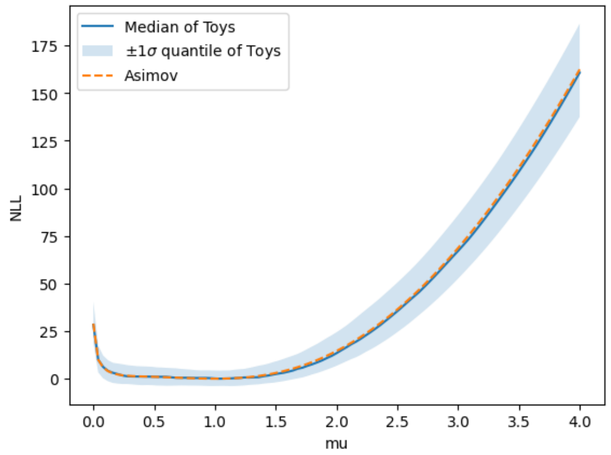

10 ifb




100 ifb



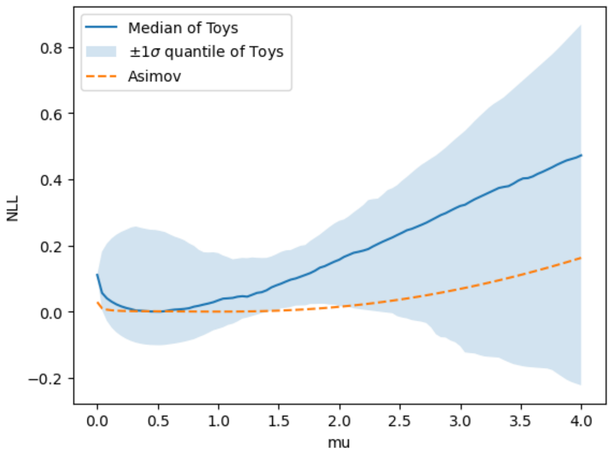

1000 ifb



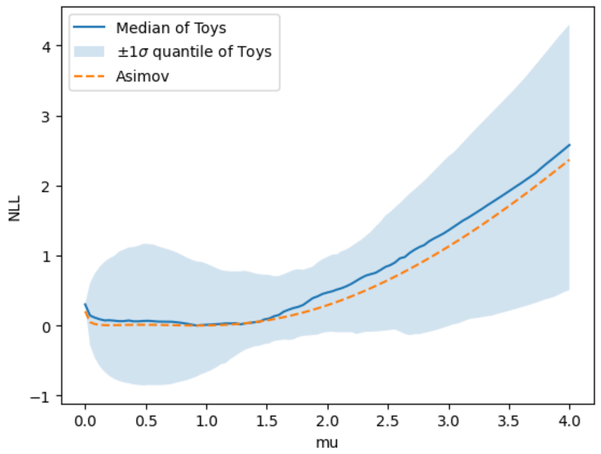

- For histogram analysis, the median of toys seems to indeed correspond to that of the single representative Asimov dataset results.

# Asimov vs. Median(Toys): NSBI

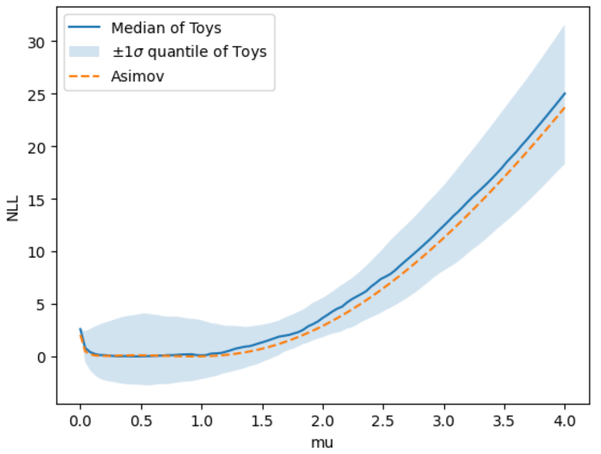
1 ifb



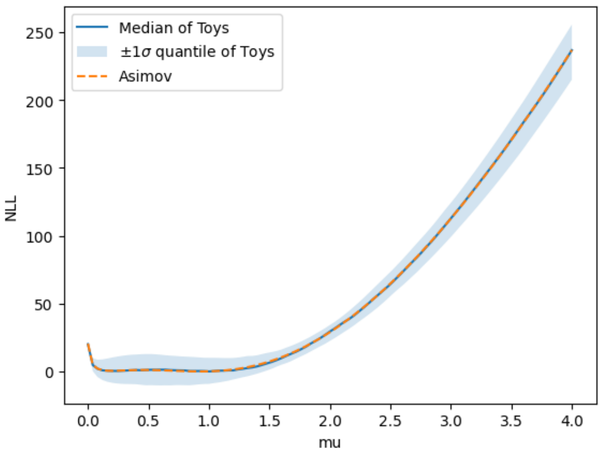

10 ifb




100 ifb




1000 ifb




- For NSBI analysis, the convergence only happens when the Poisson term takes over.
- Why does the median not correspond to the Asimov?
  - (Personal thought) the fact that as results → Poisson term the convergence is better seems to suggest that NN mis-specification could be a cause.
  - But of course, it is impossible to have a completely “un-biased” estimator with NNs (even with ensembling), in contrast to e.g. histogramming.